

CAPSTONE PROJECT CSE499B:

Final Capstone Presentation



Computer Science
and Information
Technology
Department

College of
Engineering

NetSentinel - Automated Phishing Detection and DNS Security System



By:

Mohamed Ibrahim Idris 1075626

Ameen Murtaza Siddiqui 1084960

Ahmed Mohammed Hussein 1080862

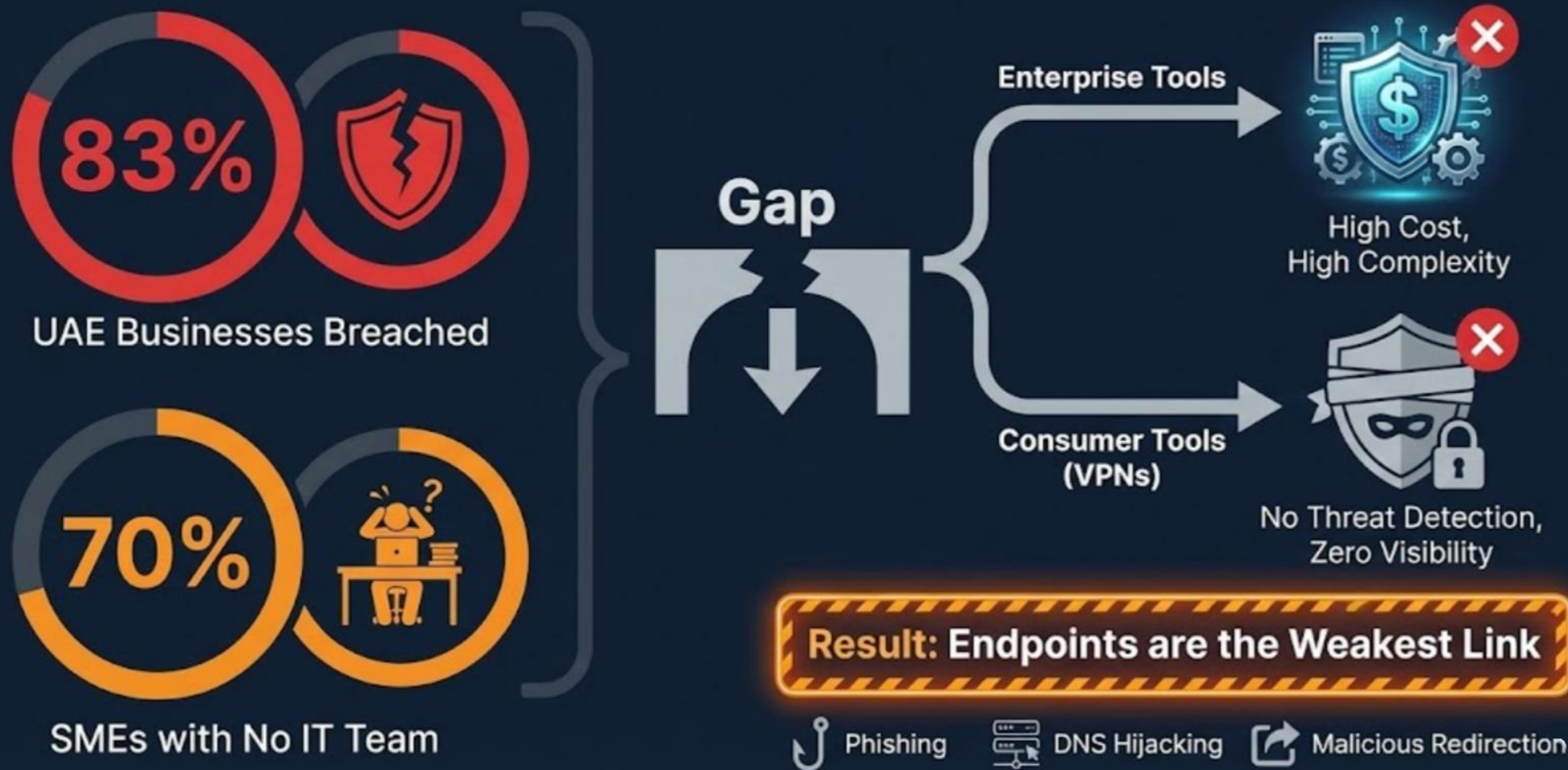
Supervised By

Dr Abid Mehmood

Outline

- **Introduction**
 - Problem Statement
- **Proposed Solution**
 - System Description
 - Design
- **Implementation**
 - Tools and Software Used
 - Working Prototype
 - AI Algorithms
- **Demo Results**
- **Project Management**
- **Security , Operational and Technical Impact**
- **Conclusion**

Problem Statement



Motivation

Pillar 1: The Context (UAE Vision)



Rapid Digital Growth
UAE is a global tech hub

High Risk
83% of businesses
report data breaches

Pillar 2: The Victim (The SME Sector)



Economic Backbone
SMEs make up 96% of
the economy

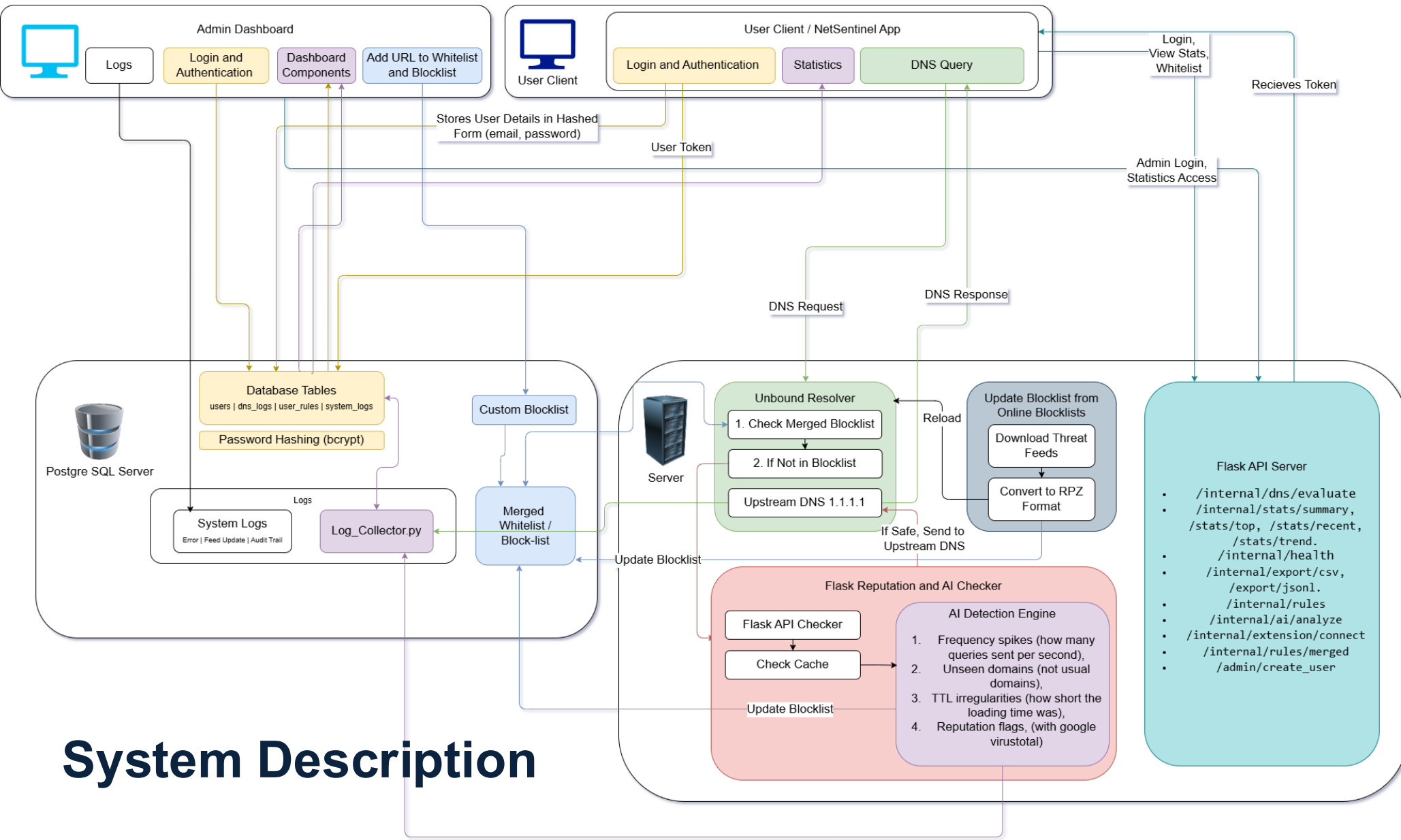
Resource Poor
70% have no dedicated
IT team

Pillar 3: The Solution

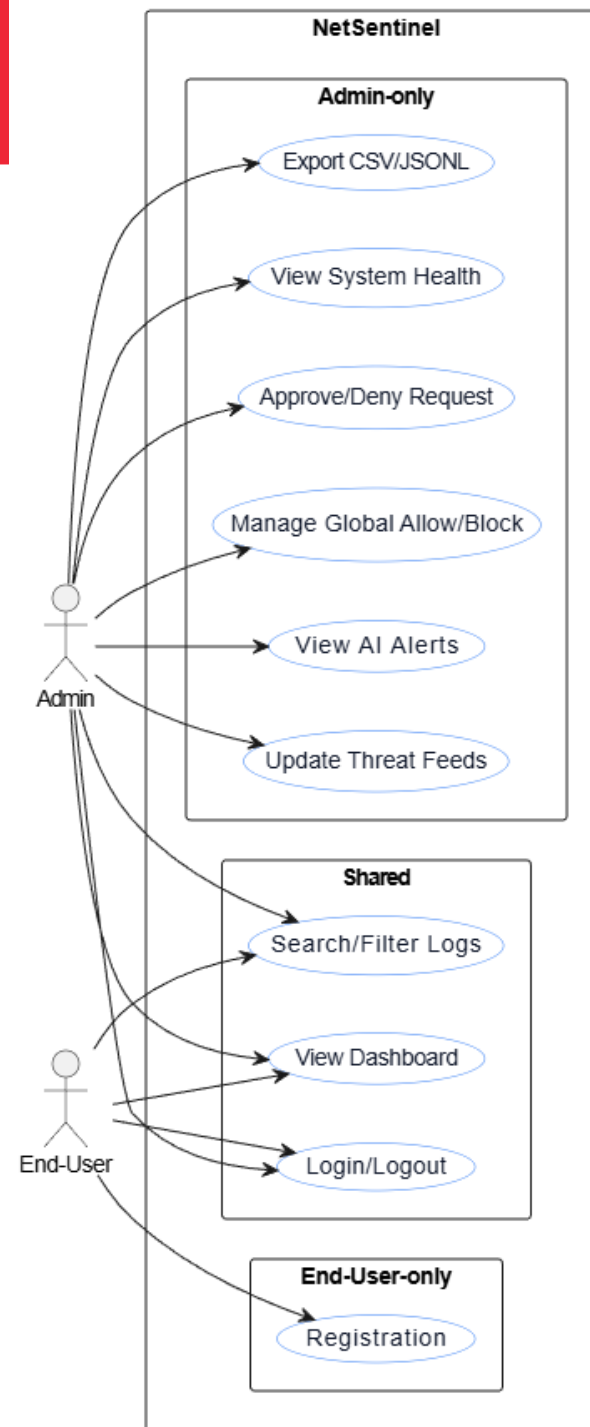


Choice between
“Expensive & Complex”
OR “Unsecure”

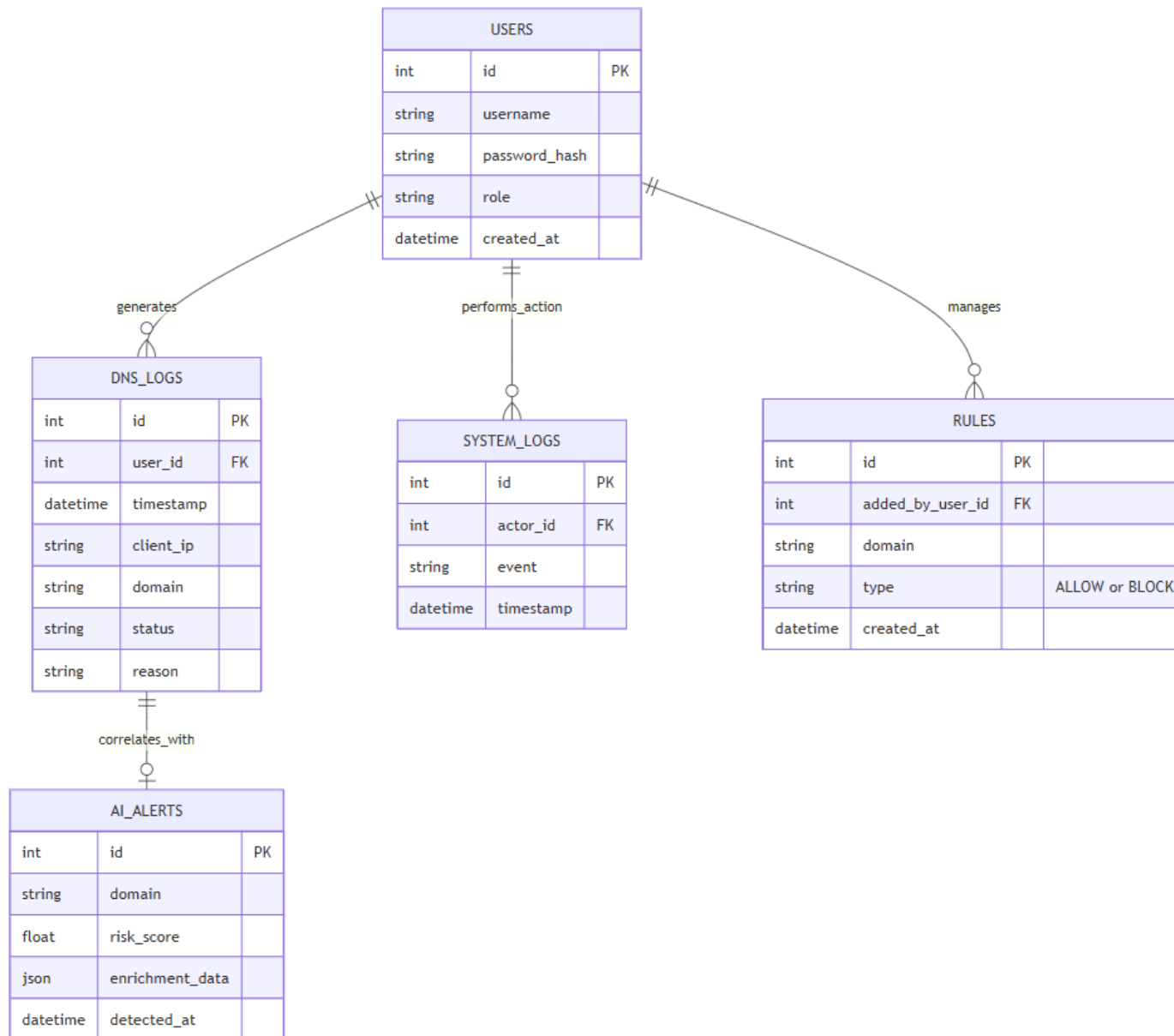
Our Mission:
Make Enterprise-Grade
security Affordable
& Automated



Design



Design

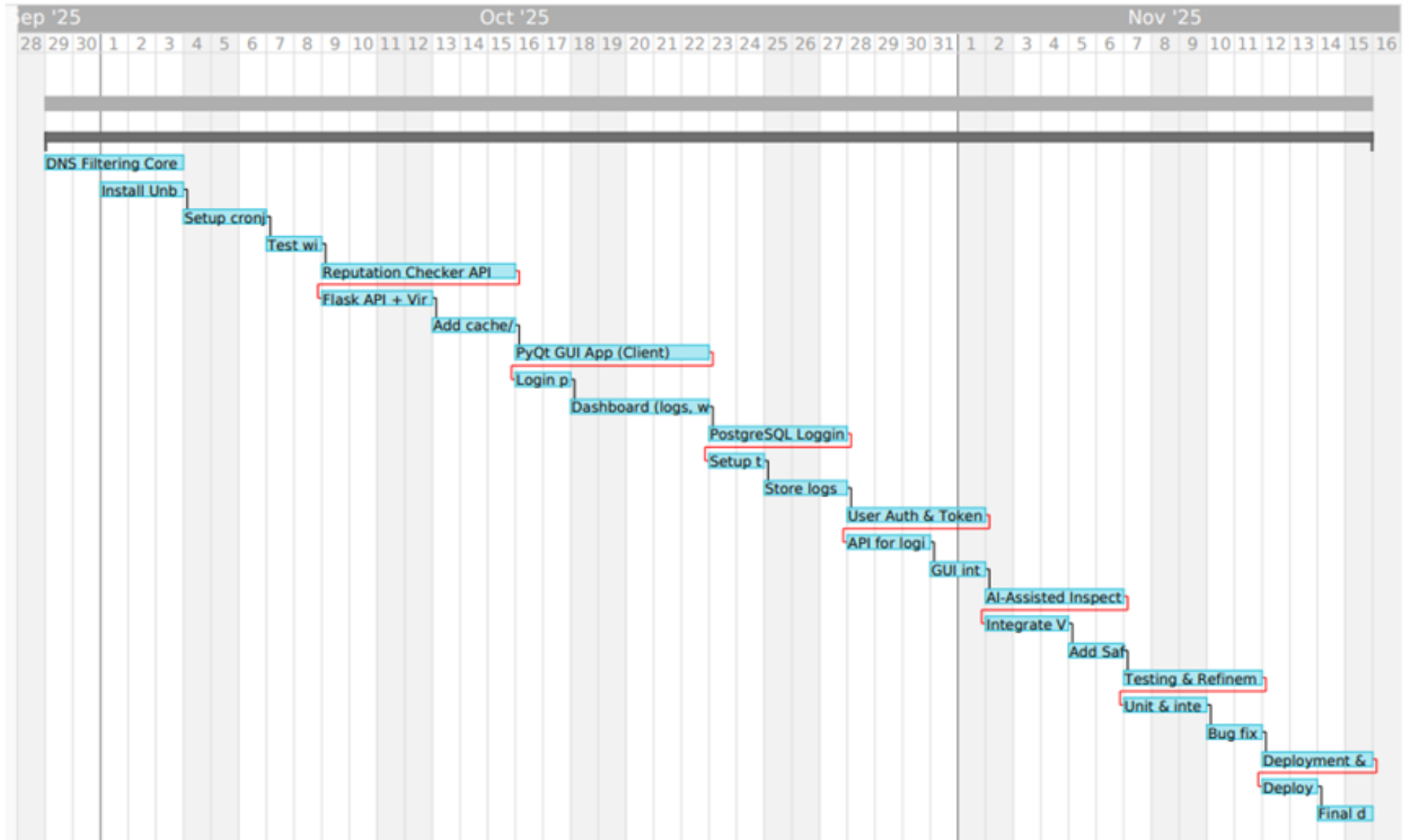


Software and Tools Used



Project Plan – Cost Estimation

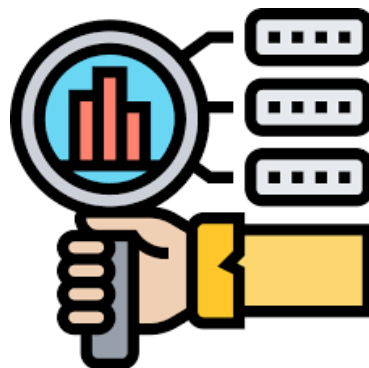
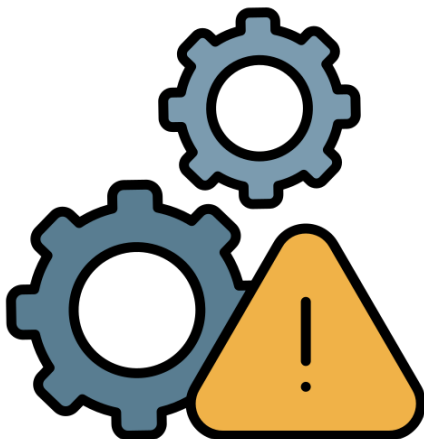
S.No	Software/Service	License/Plan	Cost (AED)
1	Ubuntu Server LTS	Open-source	0
2	Unbound (recursive DNS)	Open-source	0
3	PostgreSQL	Open-source	0
4	Nginx + Flask + systemd	Open-source	0
5	Chart.js (dashboard)	Open-source	0
6	VirusTotal API	Free tier	0
7	AlienVault OTX / AbuseIPDB	Free tier	0
8	SMTP relay / Telegram notifications	Existing / Free	0
9	Domain registration	1 year	40
10	Object storage backups (100GB)	1 year	84
Total recurring (OpEx)			124 AED / year



Security , Operational and Technical Impact



operational impact



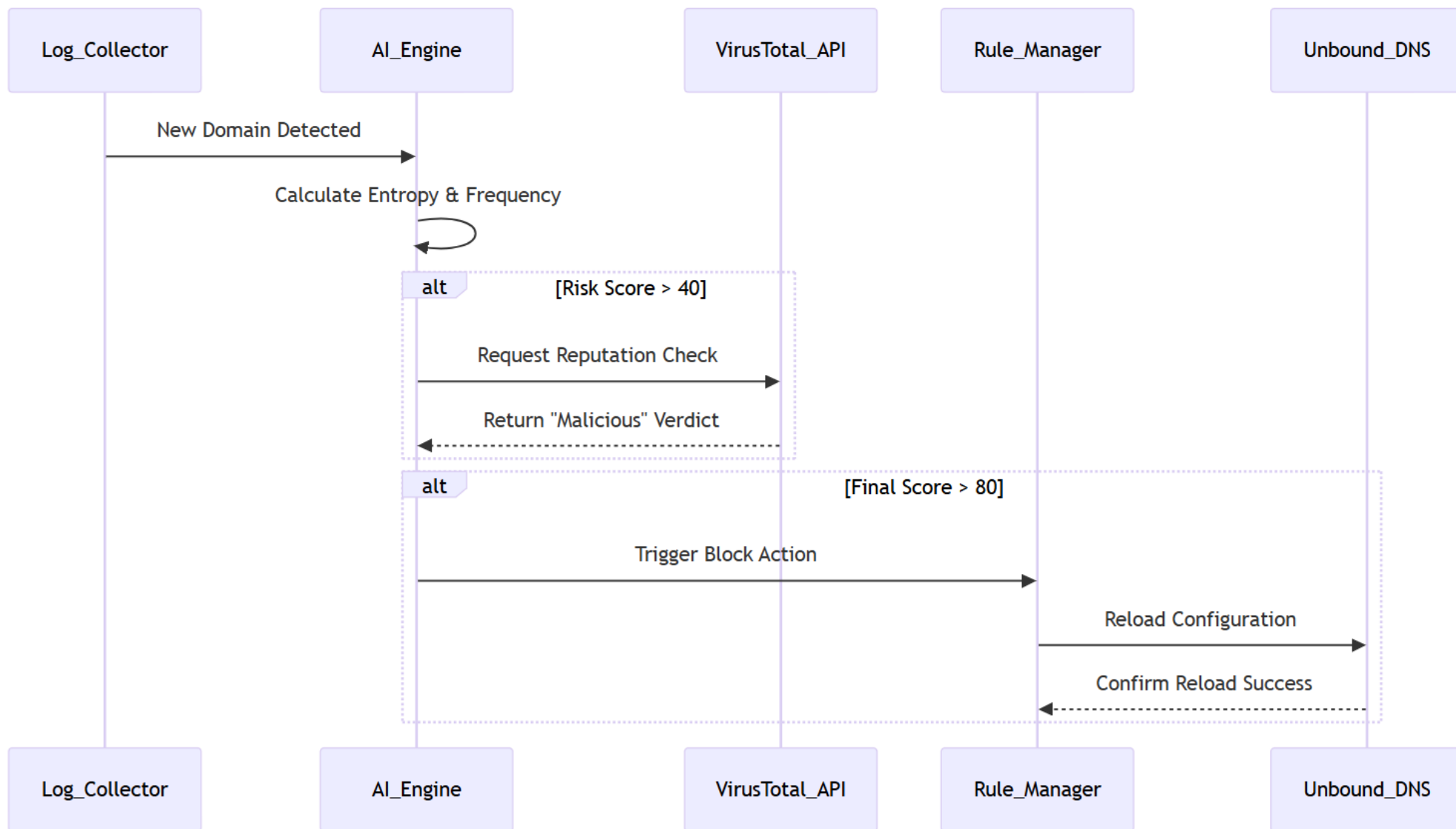
Business Impact



Technical Impact



AI Algorithm



AI Working Prototype

Domain Details

WHOIS Result

Domain Name: z-m-gateway.facebook.com
IP Address: 157.240.15.40
Registrar: RegistrarSafe, LLC
Created On: 1997-03-29 05:00:00+00:00
ASN: 32934 (US)
VT Category:
Malicious Reports: 0

NetSentinel Report

Summary: Domain name structure appears normal.

Mapped MITRE Technique: —

System Verdict: allowed

NetSentinel AI Report

AI Verdict: allowed
AI Reasoning: VT_mal=0, age=9999d, entropy=0.11, qfreq=21; mapped=T1595 Active Scanning
MITRE Mapping: —
Risk Score: 1.9%
Timestamp: 2025-11-07T15:42:37.706669Z

Close

AI Command Center

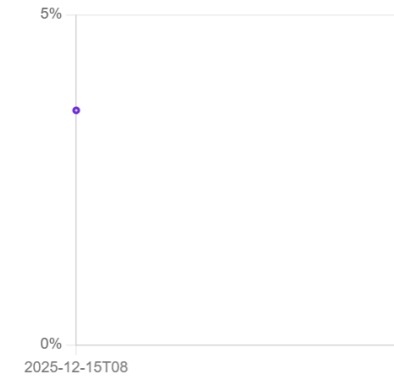
Autonomous Detection | Adaptive Defense | Real-Time Insight

AI Detected Alerts

☒ Show low-risk alerts

Time (UTC)	Domain	Risk Score	VT Category	Reason
2025-12-15T08:19:13.372942	g.whatsapp.net	2.1%	—	VT_mal=0, age=9999d, entropy=0.39, qfreq=15; mapped=T1595 Active Scanning
2025-12-15T08:19:02.527640	www.google.com	1.2%	—	VT_mal=0, age=9999d, entropy=0.18, qfreq=10; mapped=T15

AI Threat Trend



Threat Level (24 h)



Export AI Logs (CSV)

Generate a CSV summary of AI-detected domains for different time ranges.

Download CSV (1h)

Download CSV (6h)

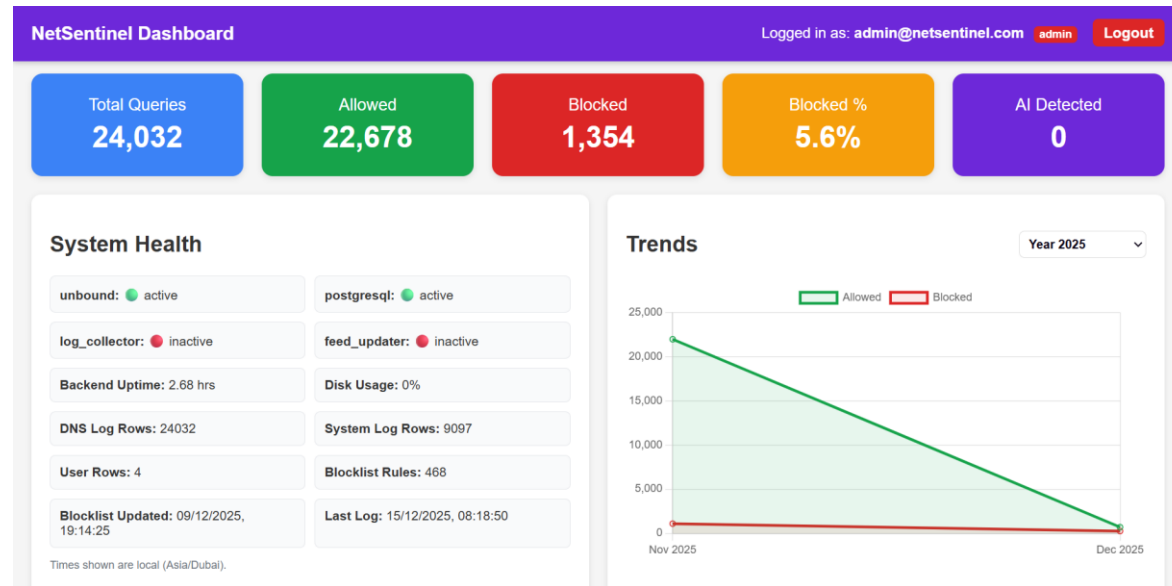
Download CSV (12h)

Download CSV (24h)

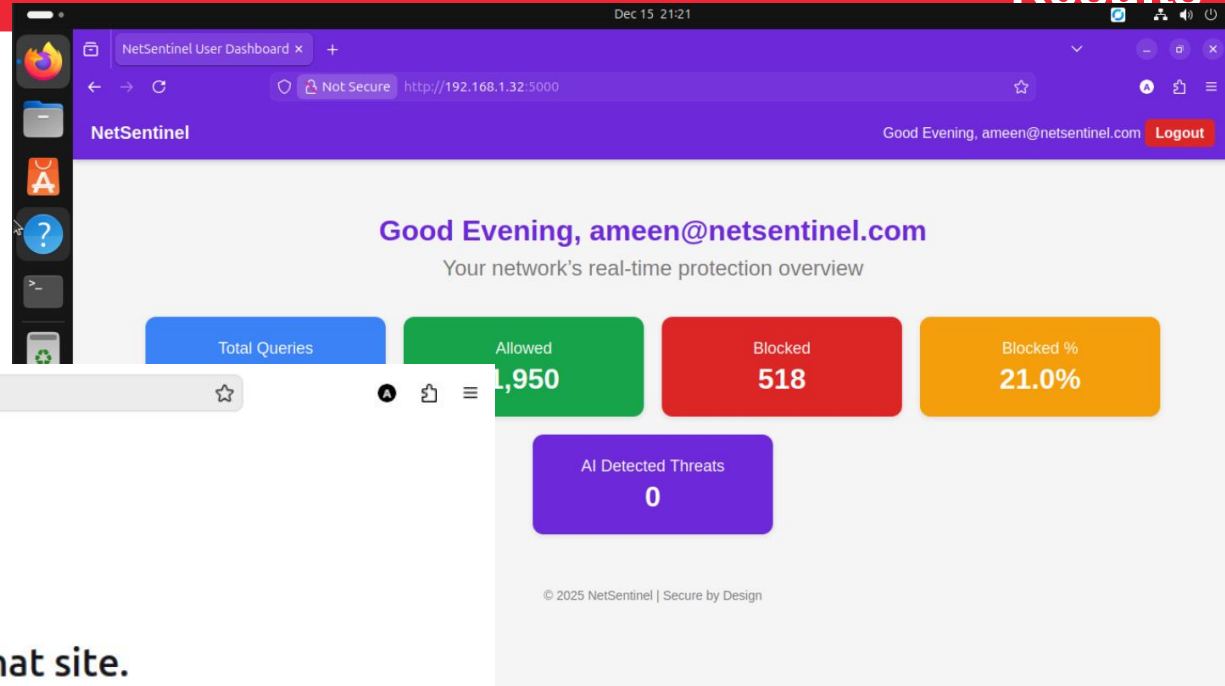
AI Settings

☐ Auto-block high-risk

Working Prototype and Demo



Results – Manual DNS Configured



Hmm. We're having trouble finding that site.

We can't connect to the server at facebook.com.

If you entered the right address, you can:

- Try again later
- Check your network connection
- Check that Firefox has permission to access the web (you might be connected but behind a firewall)

Try Again

[illegible]

Results – Desktop App

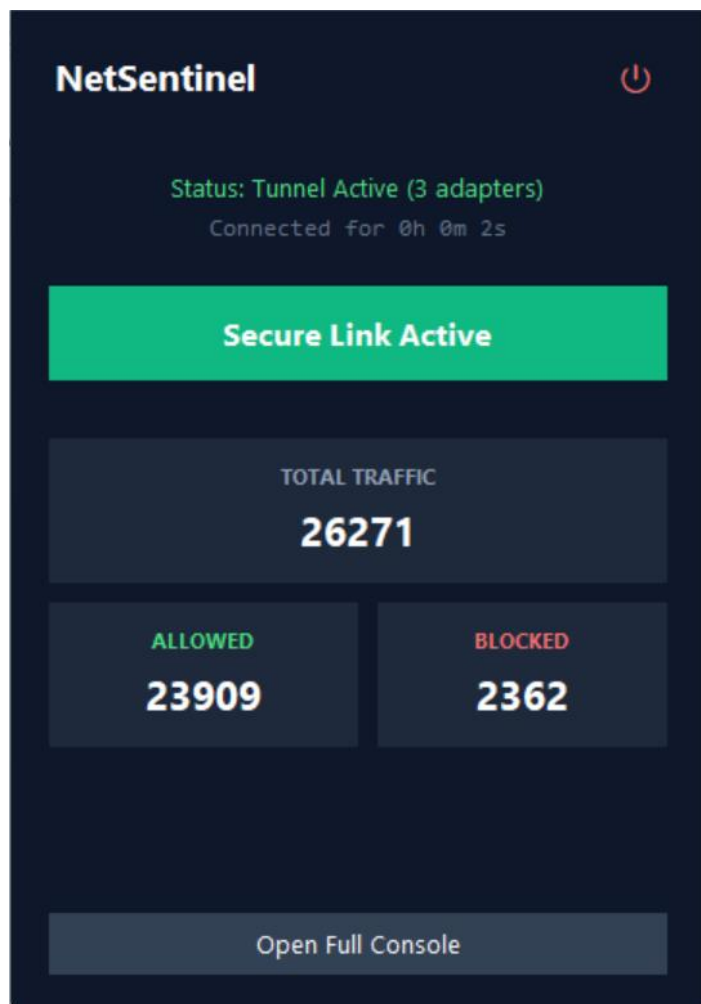


This site can't be reached

www.facebook.com's DNS address could not be found. Diagnosing the problem.

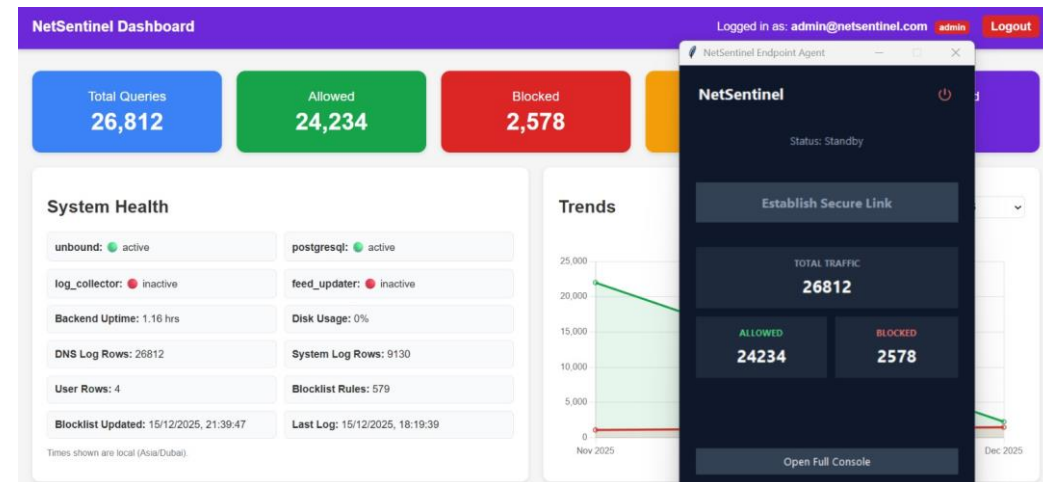
DNS PROBE POSSIBLE

Reload

[illegible]

Conclusion

- NetSentinel provides affordable, intelligent DNS-level protection against modern cyber threats.
- The system improves security visibility, operational efficiency, and incident response.
- Its scalable and lightweight design makes it suitable for SMEs and distributed environments.



NetSentinel - Automated Phishing Detection and DNS Security System

Thank You
Questions?

